

BUSINESS ANALYTICS PROJECT

SUPERSTORE BUSINESS ANALYTICS PROJECT

Student Name: Khushi Patel

Roll Number: T032

Class: TY B.Sc. IT

Division: A

Subject: Applied Business Analytics Practical

Academic Year: 2026–27

College Name: SHETH L.U.J. and Sir M.V.College of Arts ,Science and Commerce

Department Name: Bscit

Faculty Name: Chetna Tanavade

TABLE OF CONTENTS

1. Title Page
2. Table of Contents
3. Introduction
4. Problem Statement
5. Objectives
6. Business Scenario / Domain Description
7. Dataset Description
8. Data Collection
9. Data Preprocessing
10. Exploratory Data Analysis
11. Descriptive Statistical Analysis
12. Distribution Analysis
13. Correlation Analysis
14. Regression Analysis
15. Business Intelligence Dashboard
16. Probability and Probability Distributions
17. Decision Making Under Uncertainty
18. Sampling and Sampling Distribution
19. Confidence Interval Estimation
20. Bootstrapping
21. Hypothesis Testing
22. A/B Testing
23. Time Series Analysis
24. Forecast Accuracy
25. Optimization Modeling
26. Simulation Modeling
27. ANOVA
28. Experimental Design
29. Factorial Experiment
30. Business Insights and Findings
31. Business Recommendations
32. Conclusion
33. Tools and Technologies
34. Appendix

1. INTRODUCTION

Business Analytics is the process of using data, statistical methods, visualization and analytical techniques to support business decision-making.

The **Superstore Business Analytics Project** analyzes retail sales data to understand sales performance, profit, customers, products, regions, discounts and future sales trends.

Python is used to perform data preprocessing, exploratory data analysis, descriptive statistics, probability analysis, correlation, regression, hypothesis testing, forecasting, optimization, simulation and ANOVA.

The project demonstrates how Business Analytics techniques can be applied to a real-world retail business scenario.

2. PROBLEM STATEMENT

A retail organization generates a large amount of sales and customer data. It is difficult to manually identify important patterns in sales, profit, customer segments, product categories, regional performance and future demand. The problem is to analyze the available retail dataset and extract useful business information that can help management make better decisions related to:

- Sales performance
- Profitability
- Product categories
- Customer segments
- Regional performance
- Discounts
- Marketing
- Inventory
- Future sales forecasting
- Resource allocation

3. OBJECTIVES

The main objectives of this project are:

1. To understand and preprocess retail sales data.
2. To perform Exploratory Data Analysis (EDA).
3. To calculate descriptive statistics.
4. To study data distributions.
5. To identify relationships between business variables.
6. To perform correlation and regression analysis.
7. To analyze sales and profit using charts.
8. To calculate business probabilities.
9. To perform sampling and confidence interval estimation.
10. To apply bootstrapping techniques.
11. To perform hypothesis testing.
12. To demonstrate A/B testing.
13. To analyze sales trends using time series.
14. To forecast future sales.
15. To measure forecast accuracy.
16. To demonstrate optimization and simulation.
17. To perform ANOVA.
18. To demonstrate experimental and factorial design.
19. To identify important business insights.
20. To provide data-driven business recommendations.

4. BUSINESS SCENARIO / DOMAIN DESCRIPTION

Domain: Retail / E-Commerce

The project represents a retail business that sells different products to customers across different regions. The business contains information about:

- Customers
- Orders
- Products
- Categories
- Sub-categories

- Sales
- Quantity
- Discounts
- Profit
- Regions
- Customer segments
- Shipping
- Order dates

Business analytics can help the organization understand which products and regions generate more sales and profit and how discounts and other factors affect business performance.

5. DATASET DESCRIPTION

The dataset used in this project is: Superstore Business Analytics Project Dataset .

The dataset contains **10,000 records and 24 columns**.

Important attributes :

Column	Description
Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date when order was placed
Ship Date	Date when order was shipped
Ship Mode	Shipping method
Customer ID	Customer identifier
Customer Name	Customer name
Segment	Customer segment
Country	Country
City	Customer city
State	Customer state
Postal Code	Postal code
Region	Business region
Product ID	Product identifier
Category	Product category
Sub-Category	Product sub-category

Column	Description
Product Name	Product name
Sales	Sales amount
Quantity	Quantity sold
Discount	Discount provided
Profit	Profit generated
Order Month	Month of order
Profit Margin	Profit percentage
Shipping Days	Number of shipping days

6. DATA COLLECTION

The dataset was prepared for the project in CSV format.

The file used in the project is: **Superstore_Business_Analytics_Project.csv**

The dataset contains retail transaction information required to perform different Business Analytics techniques.

The CSV file was imported into Python using the Pandas library.

CODE: -

```
df = pd.read_csv("Superstore_Business_Analytics_Project.csv")
```

OUTPUT :-

```
DATASET LOADED SUCCESSFULLY
Rows: 10000
Columns: 24
```

7. DATA PREPROCESSING

Data preprocessing was performed before analysis to improve the quality and consistency of the dataset.

Steps performed:

1. Loaded the CSV dataset.
2. Checked dataset dimensions.
3. Checked column names.
4. Checked missing values.
5. Checked duplicate records.
6. Converted Order Date into datetime format.
7. Converted Ship Date into datetime format.
8. Calculated Shipping Days.
9. Created Order Month.
10. Calculated Profit Margin.
11. Removed duplicate records.

Important formulas

Shipping Days: $\text{Shipping Days} = \text{Ship Date} - \text{Order Date}$

Profit Margin: $\text{Profit Margin} = (\text{Profit} / \text{Sales}) \times 100$

Screenshot: Insert the output of preprocessing, missing-value check and duplicate check.

8. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed to understand the structure, characteristics and patterns present in the dataset.

EDA activities performed:

- Displayed first five records.
- Examined dataset information.
- Examined dataset shape.
- Checked missing values.
- Checked duplicate records.
- Calculated basic statistics.
- Identified unique categories.
- Identified regions.
- Identified customer segments.
- Counted customers and orders.
- Analyzed sales by category.
- Analyzed profit by region.
- Created charts and visualizations.

Example commands

```
df.head()
df.describe()
df.info()
df.shape
df["Category"].unique()
df["Region"].unique()
df["Segment"].unique()
```

EDA Visualizations

- Sales by Category
- Profit by Region
- Sales Distribution
- Profit Distribution
- Correlation Heatmap
- Monthly Sales Trend

EXPLORATORY DATA ANALYSIS

=====

Basic Statistics:

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings...](#)

	Row ID	Order Date	Ship Date	Postal Code	Sales	Quantity	Discount	Profit	Profit Margin
count	10000.00000	10000	10000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000
mean	5000.50000	2016-01-08 22:26:41.280000	2016-01-11 20:38:58.560000	55110.562300	414.856884	5.002000	0.158795	47.867590	10.644171
min	1.00000	2014-01-01 00:00:00	2014-01-02 00:00:00	10003.000000	2.670000	1.000000	0.000000	-360.850000	-17.994646
25%	2500.75000	2015-01-16 00:00:00	2015-01-18 00:00:00	32814.750000	121.415000	3.000000	0.050000	7.100000	5.752535
50%	5000.50000	2016-01-16 00:00:00	2016-01-19 00:00:00	55605.500000	272.995000	5.000000	0.150000	26.010000	11.485461
75%	7500.25000	2017-01-07 00:00:00	2017-01-11 00:00:00	77254.000000	558.580000	7.000000	0.200000	65.200000	17.715890
max	10000.00000	2017-12-31 00:00:00	2018-01-04 00:00:00	99986.000000	3616.900000	9.000000	0.500000	748.800000	23.997750
std	2886.89568	NaN	NaN	25930.076019	426.151968	2.574191	0.129954	72.066321	9.012908

```
Unique Categories:
['Technology' 'Office Supplies' 'Furniture']

Unique Regions:
['Central' 'South' 'East' 'West']

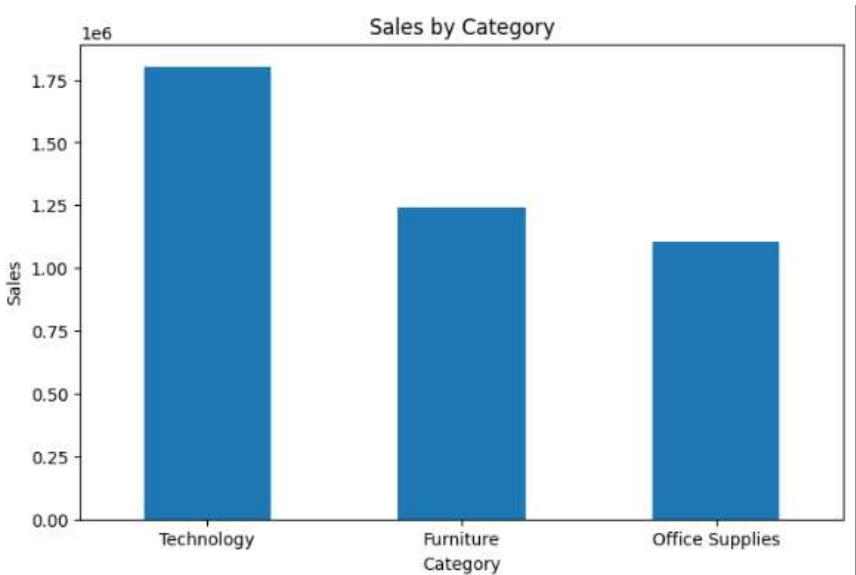
Unique Segments:
['Corporate' 'Consumer' 'Home Office']

Number of Customers:
2897

Number of Orders:
10000

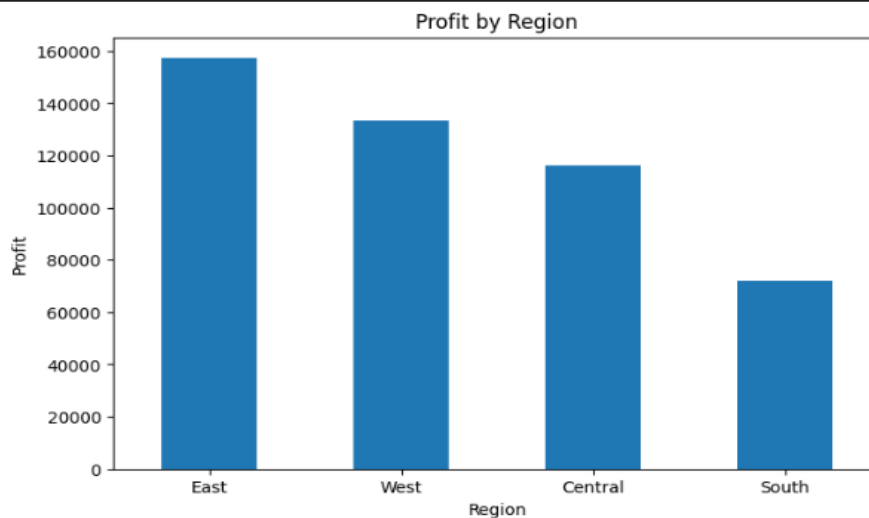
SALES BY CATEGORY

Category
Technology      1801620.31
Furniture       1243040.28
Office Supplies 1103908.25
Name: Sales, dtype: float64
```



PROFIT BY REGION

```
Region
East      157303.78
West      133185.80
Central    116088.36
South      72097.96
Name: Profit, dtype: float64
```



9. DESCRIPTIVE STATISTICAL ANALYSIS

Descriptive statistics were used to summarize the numerical data.

The following measures were calculated:

- Count
- Mean
- Median
- Standard deviation
- Minimum
- Maximum

The analysis was performed for:

- Sales
- Quantity
- Discount
- Profit
- Shipping Days
- Profit Margin

Code

```
df[
    ["Sales", "Quantity", "Discount",
     "Profit", "Shipping Days", "Profit Margin"]
].agg([
    "count", "mean", "median",
    "std", "min", "max"
])
```

Purpose

Descriptive statistics provide a basic understanding of the central tendency and variation in the dataset.

DESCRIPTIVE STATISTICS

	Sales	Quantity	Discount	Profit	Shipping Days	Profit Margin
count	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000
mean	414.856884	5.002000	0.158795	47.867590	2.925200	10.644171
median	272.995000	5.000000	0.150000	26.010000	3.000000	11.485461
std	426.151968	2.574191	0.129954	72.066321	1.258953	9.012908
min	2.670000	1.000000	0.000000	-360.850000	1.000000	-17.994646
max	3616.900000	9.000000	0.500000	748.800000	6.000000	23.997750

10. DISTRIBUTION ANALYSIS

Distribution analysis was performed to understand how numerical variables are spread across the dataset. Sales and Profit distributions were visualized using histograms.

Techniques used:

- Histogram
- Skewness

Code

```
print(df["Sales"].skew())  
print(df["Profit"].skew())
```

Histograms were created for:

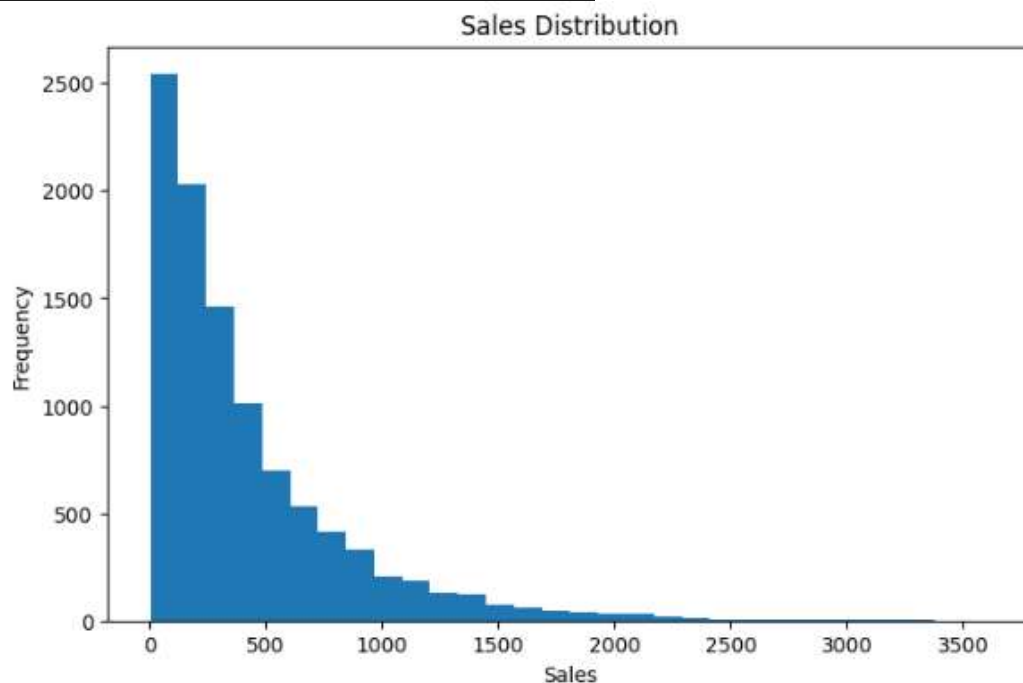
- Sales
- Profit

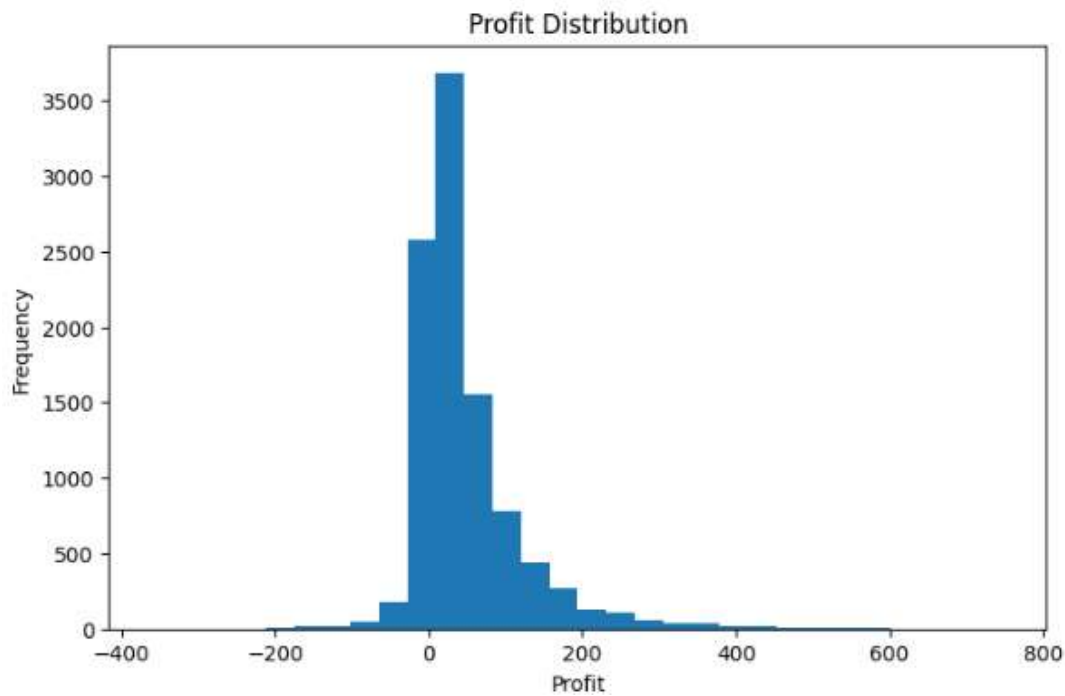
Purpose

Distribution analysis helps understand whether values are concentrated around particular ranges and whether the data is symmetric or skewed.

DISTRIBUTION ANALYSIS

```
Sales Skewness: 2.078051653543283  
Profit Skewness: 2.4996239531917523
```





11. CORRELATION ANALYSIS

Correlation analysis was performed to identify relationships between numerical variables.

Variables analyzed:

- Sales
- Quantity
- Discount
- Profit
- Shipping Days
- Profit Margin

A correlation matrix and heatmap were created.

Code

```
correlation = df[
    ["Sales", "Quantity", "Discount",
     "Profit", "Shipping Days",
     "Profit Margin"]
].corr()
```

Visualization

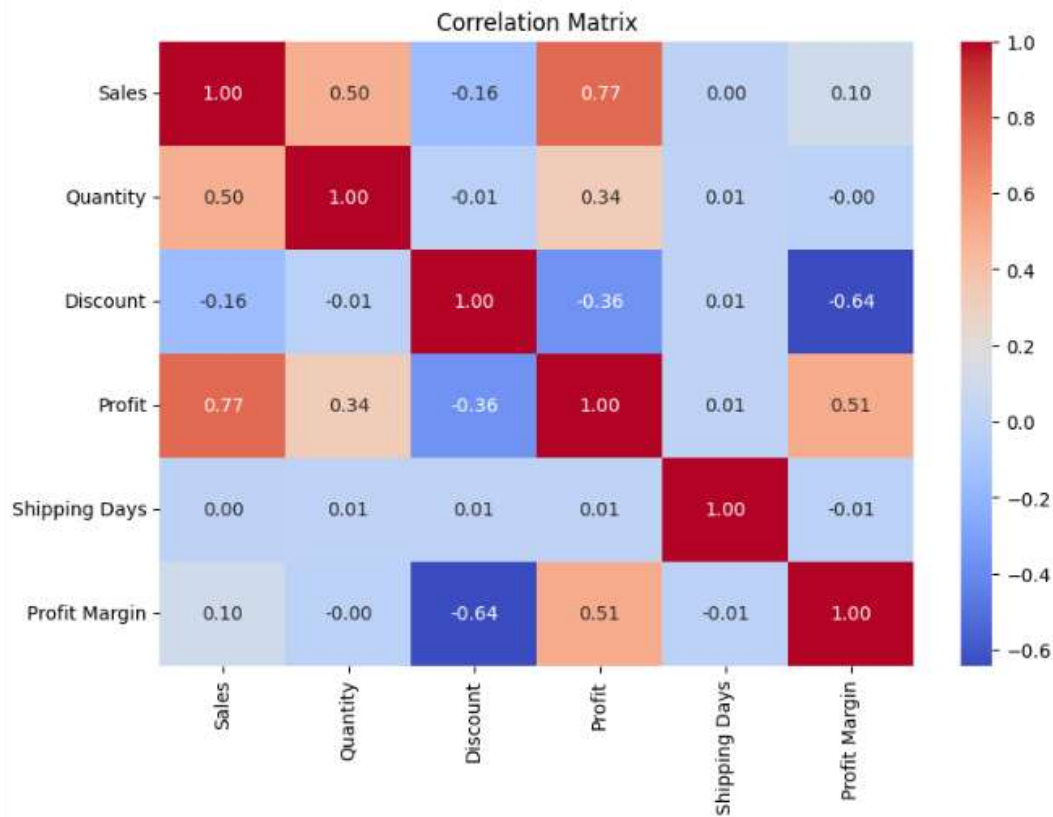
A heatmap was used to display the correlation coefficients.

Interpretation

Correlation values range from **-1 to +1**.

- Positive value → variables move in the same direction.
- Negative value → variables move in opposite directions.
- Value near 0 → weak linear relationship.

CORRELATION ANALYSIS						
	Sales	Quantity	Discount	Profit	Shipping Days	Profit Margin
Sales	1.000000	0.497440	-0.156189	0.765198	0.001322	0.096590
Quantity	0.497440	1.000000	-0.010157	0.339456	0.010292	-0.000973
Discount	-0.156189	-0.010157	1.000000	-0.361995	0.006497	-0.641436
Profit	0.765198	0.339456	-0.361995	1.000000	0.008355	0.512508
Shipping Days	0.001322	0.010292	0.006497	0.008355	1.000000	-0.006606
Profit Margin	0.096590	-0.000973	-0.641436	0.512508	-0.006606	1.000000



12. REGRESSION ANALYSIS

Regression analysis was performed to understand the relationship between business variables.

Simple Regression

A simple regression model was created using:

Independent Variable: Sales

Dependent Variable: Profit

```
X = sm.add_constant(df["Sales"])
y = df["Profit"]
```

```
model = sm.OLS(y, X).fit()
```

Multiple Regression

Multiple regression was performed using:

- Sales
- Quantity
- Discount

as independent variables and Profit as the dependent variable.

Purpose

Regression analysis helps understand how changes in sales, quantity and discount are associated with profit.

REGRESSION ANALYSIS

OLS Regression Results

```

Dep. Variable:          Profit    R-squared:                0.586
Model:                  OLS       Adj. R-squared:           0.585
Method:                 Least Squares   F-statistic:             1.412e+04
Date:                   Sat, 03 Oct 2026   Prob (F-statistic):       0.00
Time:                   19:16:34    Log-Likelihood:          -52561.
No. Observations:       10000      AIC:                     1.051e+05
Df Residuals:           9998      BIC:                     1.051e+05
Df Model:                1
Covariance Type:        nonrobust
  
```

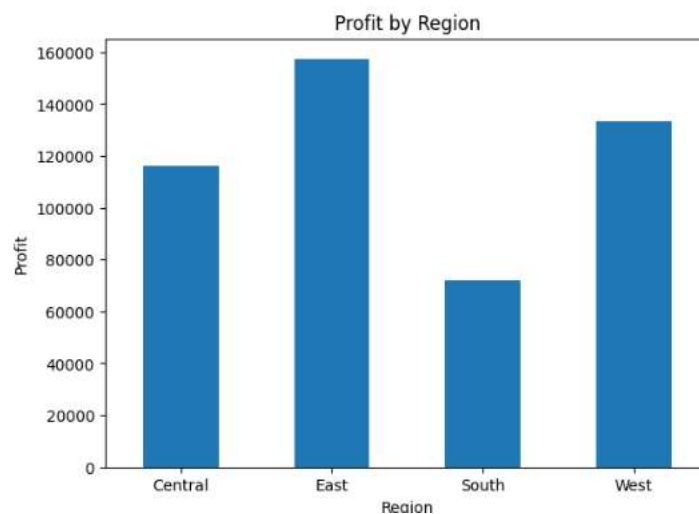
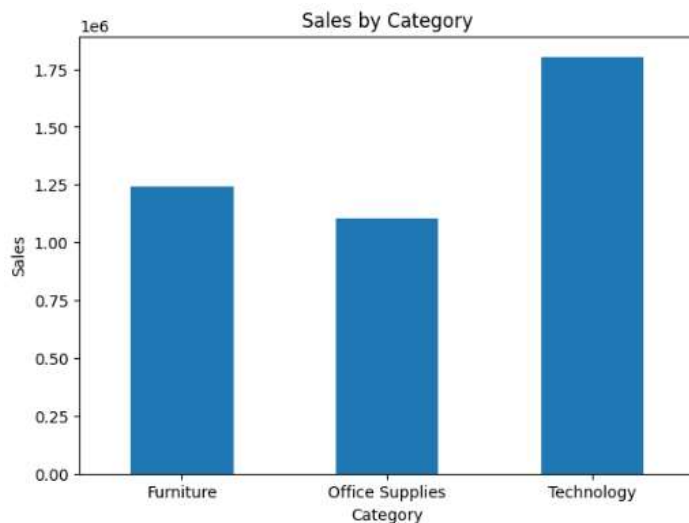
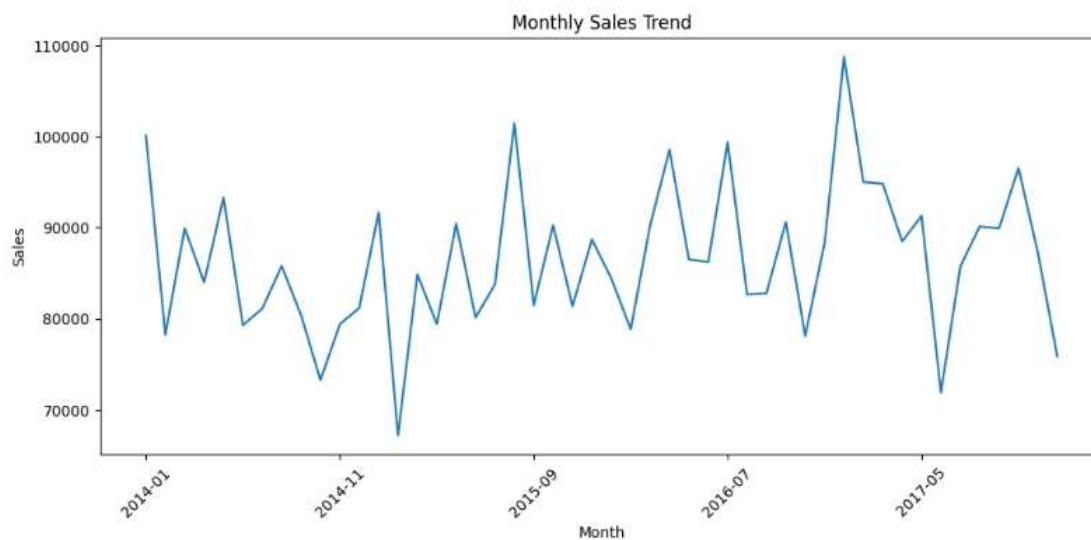
	coef	std err	t	P> t	[0.025	0.975]
const	-5.8158	0.648	-8.981	0.000	-7.085	-4.546
Sales	0.1294	0.001	118.846	0.000	0.127	0.132

```

Omnibus:                3541.890    Durbin-Watson:           2.010
Prob(Omnibus):           0.000      Jarque-Bera (JB):        69664.556
Skew:                    -1.206     Prob(JB):                0.00
Kurtosis:                15.703     Cond. No.                 830.
  
```

```

...
Total Profit: 478675.9
Total Quantity: 50020
Total Orders: 10000
Total Customers: 2897
  
```



13. BUSINESS INTELLIGENCE DASHBOARD

A Business Intelligence dashboard can be created using the analyzed dataset.

Recommended KPI Cards

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Total Quantity
- Profit Margin
- Average Order Value

Recommended Charts

1. Monthly Sales Trend
2. Sales by Category
3. Profit by Region
4. Sales by Segment
5. Sales by Sub-category
6. Discount vs Profit
7. Profit by Discount Band

Recommended Filters

- Category
- Region
- Segment
- Ship Mode
- Order Date
- Sub-category

Purpose

The dashboard provides a visual summary of business performance and allows management to understand important trends quickly.

SUPERSTORE BUSINESS ANALYTICS DASHBOARD



14. PROBABILITY AND PROBABILITY DISTRIBUTIONS

Probability analysis was performed using the transaction data.

The following probabilities were calculated:

Probability of Profit

$P(\text{Profit}) =$
Number of transactions with Profit > 0
/
Total transactions

Probability of Loss

$P(\text{Loss}) =$
Number of transactions with Profit < 0
/
Total transactions

Probability of High Discount

$P(\text{Discount} > 20\%) =$
Number of transactions with Discount > 20%
/
Total transactions

Purpose

Probability helps estimate the likelihood of different business outcomes.

```
PROBABILITY ANALYSIS
=====
Probability of Profit: 0.8779
Probability of Loss: 0.1221
Probability of Discount > 20%: 0.1993
```

15. DECISION MAKING UNDER UNCERTAINTY

Decision-making under uncertainty was demonstrated using an illustrative payoff table.

Three possible decisions were considered:

- Increase Marketing
- Maintain Marketing
- Reduce Marketing

Three possible demand conditions were considered:

- High Demand
- Medium Demand
- Low Demand

Probabilities were assigned to the demand conditions and expected values were calculated.

Expected Value Formula

Expected Value =
 $\Sigma (\text{Probability} \times \text{Payoff})$

Purpose

Expected value helps compare possible business decisions when future demand is uncertain.

```
DECISION MAKING UNDER UNCERTAINTY
=====
Illustrative Payoff Table:

  Decision  High_Demand  Medium_Demand  Low_Demand
0 Increase Marketing    50000         25000        -10000
1 Maintain Marketing    35000         30000         10000
2 Reduce Marketing     15000         18000         12000

  Decision  High_Demand  Medium_Demand  Low_Demand  Expected_Value
0 Increase Marketing    50000         25000        -10000        25500.0
1 Maintain Marketing    35000         30000         10000        27500.0
2 Reduce Marketing     15000         18000         12000        15900.0

Expected values are calculated using assumed business probabilities.
```

16. SAMPLING AND SAMPLING DISTRIBUTION

Sampling was performed by repeatedly selecting samples of sales data.

Method

- Sample size = 50
- Number of samples = 1000
- Sampling was performed with replacement.
- Mean sales was calculated for each sample.

The resulting sample means were used to create a sampling distribution.

Purpose

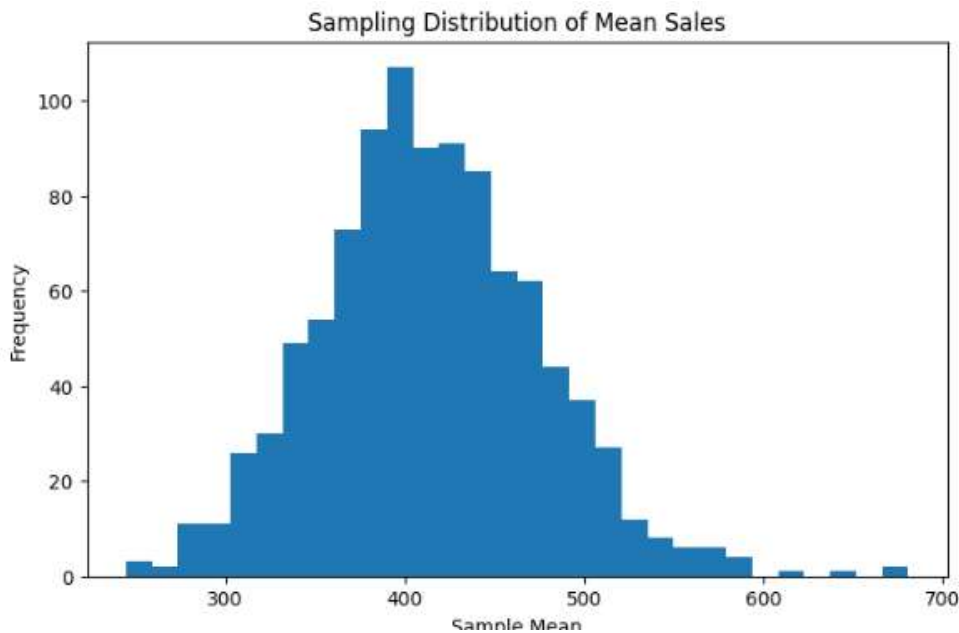
Sampling allows businesses to study a smaller portion of data and use it to understand the larger population.

SAMPLING AND SAMPLING DISTRIBUTION

Number of Samples: 1000

Mean of Sample Means: 414.39093319999995

Standard Deviation: 61.467687118405365



17. CONFIDENCE INTERVAL ESTIMATION

Confidence intervals were calculated for the mean Sales.

Confidence levels used:

- 90%
- 95%
- 99%

General formula

Confidence Interval =

Sample Mean $\pm$ Margin of Error

The t-distribution was used to calculate the confidence intervals.

Purpose

Confidence intervals provide a range within which the population mean is estimated to lie at a specified confidence level.

CONFIDENCE INTERVAL ESTIMATION

90% CI: (407.85, 421.87)

95% CI: (406.50, 423.21)

99% CI: (403.88, 425.84)

18. BOOTSTRAPPING

Bootstrapping is a resampling technique used to estimate the uncertainty of a statistic.

In this project:

- 1000 bootstrap samples were generated.
- Each sample was selected with replacement.
- The mean Sales was calculated for each bootstrap sample.
- A 95% bootstrap confidence interval was calculated.

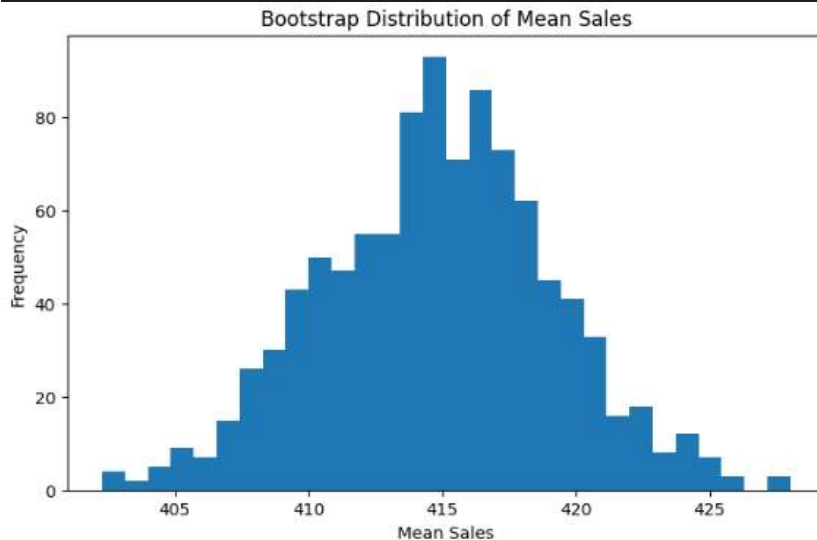
Purpose

Bootstrapping is useful when assumptions about the underlying population distribution are difficult to make.

BOOTSTRAPPING

```
Bootstrap Mean: 414.86338693799996
```

```
95% Bootstrap CI: (np.float64(406.29900365), np.float64(423.436958175))
```



19. HYPOTHESIS TESTING

A one-sample t-test was performed on Profit.

Hypotheses

Null Hypothesis (H_0): Mean Profit = 0

Alternative Hypothesis (H_1): Mean Profit $\neq$ 0

Significance Level

$\alpha = 0.05$

Decision Rule

- If p-value < 0.05 $\rightarrow$ Reject H_0 .
- If p-value $\geq$ 0.05 $\rightarrow$ Fail to reject H_0 .

Purpose

Hypothesis testing determines whether the observed result provides statistical evidence against a specified null hypothesis.

HYPOTHESIS TESTING

```
T-statistic: 66.42158117512652
```

```
P-value: 0.0
```

```
Result: Reject H0. Mean profit is significantly different from zero.
```

20. A/B TESTING

A/B testing was demonstrated using two hypothetical marketing groups.

Group A

Email Marketing

Group B

Social Media Marketing

Each group contained:

5,000 visitors

Conversions:

- Group A = 350
- Group B = 410

Conversion rates were calculated and compared using a two-proportion z-test.

Purpose

A/B testing can help a business compare two marketing strategies and determine whether their conversion rates differ statistically.

Important: These A/B test values are **simulated for demonstration**, because the original dataset does not contain randomized A/B experiment groups.

A/B TESTING

Group A Conversion Rate: 0.07

Group B Conversion Rate: 0.082

Z-statistic: -2.2641670642586305

P-value: 0.02356384145986638

The difference between the two conversion rates is statistically significant.

21. TIME SERIES ANALYSIS

Time series analysis was performed using monthly Sales data.

The dataset was grouped by **Order Month**.

A monthly sales line graph was created.

Purpose

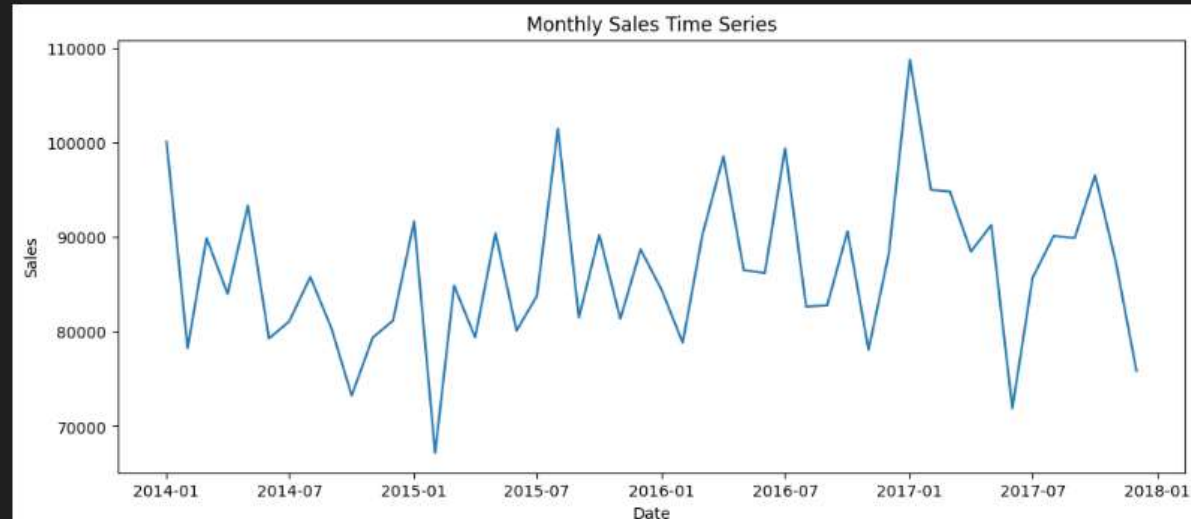
Time series analysis helps identify:

- Trends
- Changes over time
- Seasonal patterns
- Future business behavior

Visualization

A line chart was created to display monthly sales.

TIME SERIES ANALYSIS



22. FORECAST ACCURACY

Exponential Smoothing was used to forecast future sales.

The final six months were separated as a test dataset to evaluate forecast accuracy.

The following measures were calculated:

MAE

MAE = Mean Absolute Error

MSE

MSE = Mean Squared Error

RMSE

RMSE = Root Mean Squared Error

Purpose

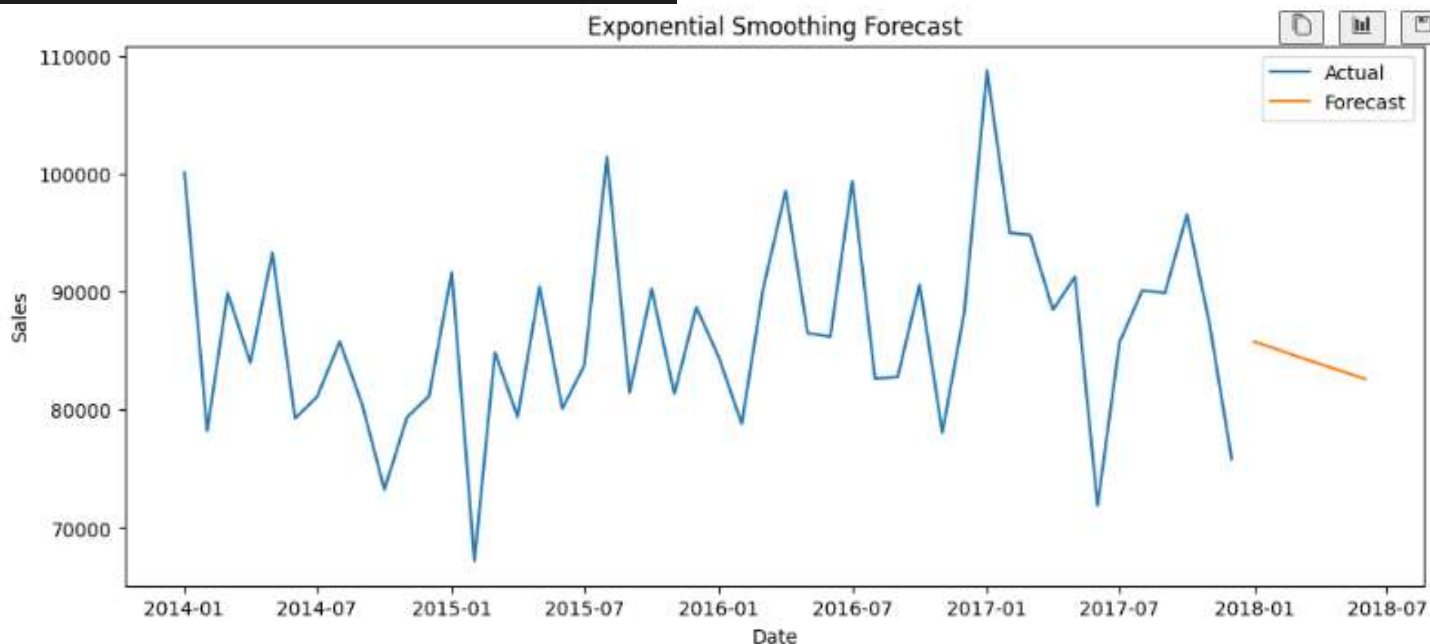
Forecast accuracy measures how close predicted sales are to actual sales.

EXPONENTIAL SMOOTHING FORECAST

Next 6 Month Forecast:

2018-01-01	85765.565451
2018-02-01	85138.976469
2018-03-01	84512.387487
2018-04-01	83885.798506
2018-05-01	83259.209524
2018-06-01	82632.620542

Freq: MS, dtype: float64



FORECAST ACCURACY

MAE:	4610.298169426959
MSE:	38546590.11260838
RMSE:	6208.590026133823

23. OPTIMIZATION MODELING

Optimization modeling was demonstrated using a linear programming model.

Three product groups were considered:

- Furniture
- Office Supplies

- Technology

The objective was to maximize estimated profit while satisfying resource constraints.

Objective

Maximize Total Profit

subject to:

- Labour constraint
- Storage constraint
- Non-negative product quantities

Purpose

Optimization helps businesses allocate limited resources efficiently.

```
OPTIMIZATION MODELING
=====
Optimization Successful: True
Optimal Units:
Furniture: 0.0
Office Supplies: 75.0
Technology: 0.0
Maximum Estimated Profit: 2250.0
```

24. SIMULATION MODELING

Simulation was used to demonstrate inventory management.

Daily demand was generated using a Poisson distribution.

Simulation Parameters

- Simulation period = 30 days
- Starting inventory = 100 units
- Average daily demand = 20 units

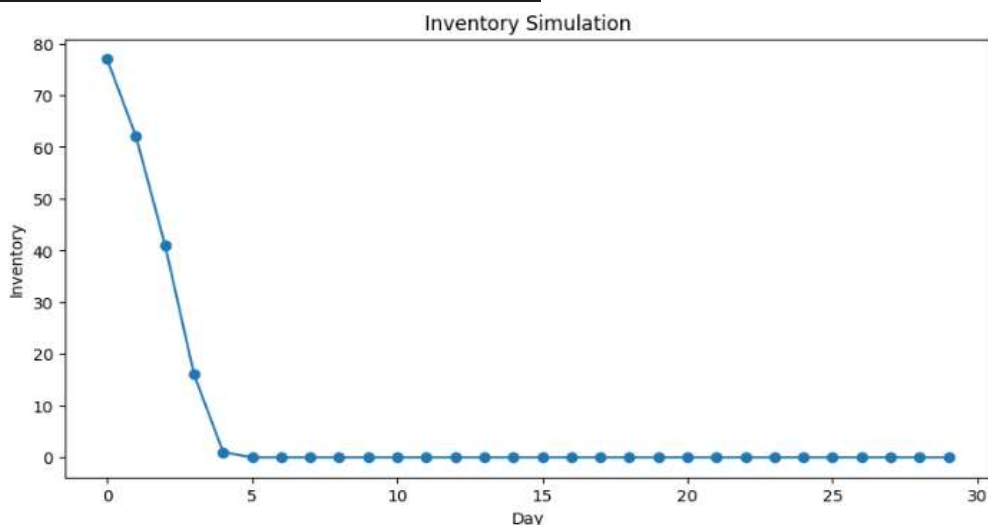
The simulation calculated:

- Daily demand
- Inventory level
- Stockout days
- Ending inventory

Purpose

Simulation helps businesses understand possible future outcomes under uncertain demand.

```
SIMULATION MODELING
=====
Total Demand: 585
Stockout Days: 25
Ending Inventory: 0
```



25. ANOVA

Analysis of Variance (ANOVA) was performed to compare Profit across different product categories.

Categories

- Furniture
- Office Supplies
- Technology

Hypotheses

H₀:

Mean profit is equal across categories.

H₁:

At least one category has a different mean profit.

Significance Level

$\alpha = 0.05$

Purpose

ANOVA determines whether there are statistically significant differences between the means of multiple groups.

ANOVA				
=====				
	sum_sq	df	F	PR(>F)
C(Category)	4.439043e+06	2.0	467.212929	1.053425e-194
Residual	4.749131e+07	9997.0	NaN	NaN
ANOVA P-value: 1.0534250036198289e-194				
There is a statistically significant difference in profit among categories.				

26. EXPERIMENTAL DESIGN

An experimental design was proposed to evaluate marketing performance.

Factor

Promotion Type

Levels

1. Email Marketing
2. Social Media Marketing

Response Variable

Sales

Experimental Units

Customers

Procedure

1. Randomly divide customers into groups.
2. Apply Email Marketing to Group A.
3. Apply Social Media Marketing to Group B.
4. Record sales.
5. Compare the results.
6. Perform statistical testing.

EXPERIMENTAL DESIGN	
=====	
Experimental Design:	
Factor:	Promotion Type
Levels:	1. Email Marketing 2. Social Media Marketing
Response Variable:	Sales
Experimental Units:	Customers
Procedure:	1. Randomly divide customers into two groups. 2. Apply Email Marketing to Group A. 3. Apply Social Media Marketing to Group B.

27. FACTORIAL EXPERIMENT

A 2×2 factorial experiment was demonstrated.

Factor 1: Promotion

- Email
- Social Media

Factor 2: Discount Level

- Low
- High

Response Variable

Sales

This creates four combinations:

Promotion Discount

Email Low

Email High

Social Media Low

Social Media High

Two-way ANOVA was used to analyze the effects of the factors and their interaction.

Purpose

Factorial experiments help study the individual and combined effects of multiple factors.

Note: The factorial observations are simulated for demonstrating the technique.

FACTORIAL EXPERIMENT					
=====					
2 × 2 Factorial Experiment:					
Output is truncated. View as a scrollable element or open in a text editor . Adjust cell output settings...					
	Promotion	Discount_Level	Sales		
0	Email	Low	549.671415		
1	Email	Low	486.173570		
2	Email	Low	564.768854		
3	Email	Low	652.302986		
4	Email	Low	476.584663		
			sum_sq	df	PR(>F)
C(Promotion)			7189.129461	1.0	0.762917
C(Discount_Level)			493.846439	1.0	0.052407
C(Promotion):C(Discount_Level)			404.733365	1.0	0.042951
Residual			716163.808084	76.0	NaN

28. BUSINESS INSIGHTS AND FINDINGS

The analysis provides several useful business insights.

Major findings

1. The dataset contains **10,000 retail transaction records**.
2. Sales performance can be compared across different product categories.
3. Regional analysis identifies differences in profitability.
4. Customer segments can be compared based on sales.
5. Product sub-categories can be analyzed to identify high-performing products.
6. Sales and profit distributions provide information about data variability.
7. Correlation analysis identifies relationships among numerical business variables.
8. Regression analysis helps examine factors associated with profit.
9. Probability analysis measures the likelihood of profit, loss and high discounts.
10. Sampling provides information about population behavior using smaller samples.
11. Confidence intervals estimate population parameters.
12. Bootstrapping provides an alternative method for estimating uncertainty.

13. Hypothesis testing provides statistical evidence for business questions.
14. Time series analysis helps understand sales trends.
15. Forecasting supports future sales planning.
16. Optimization supports efficient allocation of limited resources.
17. Simulation helps analyze uncertain inventory demand.
18. ANOVA allows comparison of profit across categories.
19. Experimental design provides a structured approach for testing marketing strategies.
20. Factorial design allows multiple factors to be studied simultaneously.

29. BUSINESS RECOMMENDATIONS

Based on the analysis, the following recommendations can be made:

1. Monitor Product Performance

Regularly analyze category and sub-category sales to identify important product groups.

2. Analyze Regional Performance

Compare sales and profit across regions and investigate areas with lower performance.

3. Control Discounts

Discounts should be monitored because excessive discounts may affect profitability.

4. Use Sales Forecasting

Forecasting can support inventory and purchasing decisions.

5. Improve Inventory Planning

Simulation can be used to understand demand uncertainty and possible stockouts.

6. Use Customer Segmentation

Marketing campaigns can be designed according to customer segments.

7. Test Marketing Campaigns

A/B testing can be used before implementing large-scale marketing campaigns.

8. Use Optimization

Optimization models can support efficient use of limited resources.

9. Monitor KPIs

A BI dashboard can provide management with regular information about sales, profit and customers.

10. Make Data-Driven Decisions

Business decisions should be supported by statistical analysis and measurable evidence.

30. CONCLUSION

The **Superstore Business Analytics Project** demonstrates the complete application of Business Analytics techniques to retail transaction data.

The project started with data loading and preprocessing and continued with Exploratory Data Analysis, descriptive statistics, distribution analysis, correlation, regression and visualization.

Advanced analytics techniques including probability, decision-making under uncertainty, sampling, confidence intervals, bootstrapping, hypothesis testing, A/B testing, time series forecasting, optimization, simulation and ANOVA were also demonstrated. The project shows how data analysis can help a retail organization understand its performance and support decisions related to sales, profit, marketing, inventory and resource allocation.

Overall, the project demonstrates the importance of using data-driven methods for effective business decision-making.

31. APPENDIX

1] Complete Python Code :

1. IMPORT LIBRARIES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
from scipy import stats
from scipy.optimize import linprog
import statsmodels.api as sm
from statsmodels.formula.api import ols
from statsmodels.tsa.holtwinters import ExponentialSmoothing
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.stats.proportion import proportions_ztest
warnings.filterwarnings("ignore")
print("="*70)
print("APPLIED BUSINESS ANALYTICS PROJECT")
print("SUPERSTORE BUSINESS ANALYTICS")
print("="*70)
```

2. LOAD DATASET

```
df =
pd.read_csv("Superstore_Business_Analytics_Project.csv")
print("\nDATASET LOADED SUCCESSFULLY")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
print("\nFirst 5 Records:")
display(df.head())
```

3. DATASET INFORMATION

```
print("\nDATASET INFORMATION")
print("="*50)
df.info()
print("\nColumn Names:")
print(df.columns.tolist())
```

4. DATA PREPROCESSING

```
print("\nDATA PREPROCESSING")
print("="*50)
```

Convert dates

```
df["Order Date"] = pd.to_datetime(df["Order Date"])
df["Ship Date"] = pd.to_datetime(df["Ship Date"])
```

Calculate shipping days

```
df["Shipping Days"] = (
    df["Ship Date"] - df["Order Date"]
).dt.days
```

Create month column

```
df["Order Month"] = df["Order
Date"].dt.to_period("M").astype(str)
```

Create profit margin

```
df["Profit Margin"] = (
    df["Profit"] / df["Sales"] * 100
)
```

Remove duplicate rows

```
df = df.drop_duplicates()
print("Preprocessing completed.")
print("Rows after preprocessing:", len(df))
```

5. MISSING VALUES

```
print("\nMISSING VALUES")
print("="*50)
missing = df.isnull().sum()
print(missing)
```

6. DUPLICATE VALUES

```
print("\nDUPLICATE RECORDS")
print("="*50)
print("Duplicate rows:", df.duplicated().sum())
```

7. EXPLORATORY DATA ANALYSIS

```
print("\nEXPLORATORY DATA ANALYSIS")
print("="*50)
print("\nBasic Statistics:")
display(df.describe())
print("\nUnique Categories:")
print(df["Category"].unique())
print("\nUnique Regions:")
print(df["Region"].unique())
print("\nUnique Segments:")
print(df["Segment"].unique())
print("\nNumber of Customers:")
print(df["Customer ID"].nunique())
print("\nNumber of Orders:")
print(df["Order ID"].nunique())
```

8. SALES BY CATEGORY

```
category_sales =
df.groupby("Category")["Sales"].sum().sort_values(
    ascending=False
)
print("\nSALES BY CATEGORY")
display(category_sales)
plt.figure(figsize=(8,5))
category_sales.plot(kind="bar")
plt.title("Sales by Category")
plt.xlabel("Category")
```

```
plt.ylabel("Sales")
plt.xticks(rotation=0)
plt.show()
```

9. PROFIT BY REGION

```
region_profit =
df.groupby("Region")["Profit"].sum().sort_values(
    ascending=False
) print("\nPROFIT BY REGION")
display(region_profit)
plt.figure(figsize=(8,5))
region_profit.plot(kind="bar")
plt.title("Profit by Region")
plt.xlabel("Region")
plt.ylabel("Profit")
plt.xticks(rotation=0)
plt.show()
```

10. DESCRIPTIVE STATISTICS

```
print("\nDESCRIPTIVE STATISTICS")
print("="*50)
statistics = df[[
    "Sales",
    "Quantity",
    "Discount",
    "Profit",
    "Shipping Days",
    "Profit Margin"
]].agg([
    "count",
    "mean",
    "median",
    "std",
    "min",
    "max"
])
display(statistics)
```

11. DISTRIBUTION ANALYSIS

```
print("\nDISTRIBUTION ANALYSIS")
print("="*50)
print("Sales Skewness:", df["Sales"].skew())
print("Profit Skewness:", df["Profit"].skew())
plt.figure(figsize=(8,5))
plt.hist(df["Sales"], bins=30)
plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()
plt.figure(figsize=(8,5))
plt.hist(df["Profit"], bins=30)
plt.title("Profit Distribution")
plt.xlabel("Profit")
plt.ylabel("Frequency")
```

```
plt.show()
```

12. CORRELATION ANALYSIS

```
print("\nCORRELATION ANALYSIS")
print("="*50)
numeric_columns = [
    "Sales",
    "Quantity",
    "Discount",
    "Profit",
    "Shipping Days",
    "Profit Margin"
]
correlation = df[numeric_columns].corr()
display(correlation)
plt.figure(figsize=(9,6))
sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)
plt.title("Correlation Matrix")
plt.show()
```

13. REGRESSION ANALYSIS

```
print("\nREGRESSION ANALYSIS")
print("="*50)
# Simple regression
X = df["Sales"]
y = df["Profit"]
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary())

# Multiple regression
print("\nMULTIPLE REGRESSION")
```

```
X_multi = df[
    ["Sales", "Quantity", "Discount"]
]
X_multi = sm.add_constant(X_multi)
model_multi = sm.OLS(
    df["Profit"],
    X_multi
).fit()
print(model_multi.summary())
```

14. BUSINESS INTELLIGENCE DASHBOARD DATA

```
print("\nBUSINESS INTELLIGENCE DASHBOARD")
print("="*50)
total_sales = df["Sales"].sum()
total_profit = df["Profit"].sum()
total_quantity = df["Quantity"].sum()
```

```
total_orders = df["Order ID"].nunique()
total_customers = df["Customer ID"].nunique()
print("Total Sales:", round(total_sales, 2))
print("Total Profit:", round(total_profit, 2))
print("Total Quantity:", total_quantity)
print("Total Orders:", total_orders)
print("Total Customers:", total_customers)
```

```
# Monthly Sales
monthly_sales = df.groupby("Order Month")["Sales"].sum()
plt.figure(figsize=(12,5))
monthly_sales.plot()
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.show()
```

```
# Category
plt.figure(figsize=(7,5))
df.groupby("Category")["Sales"].sum().plot(kind="bar")
plt.title("Sales by Category")
plt.ylabel("Sales")
plt.xticks(rotation=0)
plt.show()
```

```
# Region
plt.figure(figsize=(7,5))
df.groupby("Region")["Profit"].sum().plot(kind="bar")
plt.title("Profit by Region")
plt.ylabel("Profit")
plt.xticks(rotation=0)
plt.show()
```

15. PROBABILITY

```
print("\nPROBABILITY ANALYSIS")
print("="*50)
# Probability of profit
prob_profit = (df["Profit"] > 0).mean()
# Probability of loss
prob_loss = (df["Profit"] < 0).mean()
# Probability of high discount
prob_high_discount = (df["Discount"] > 0.20).mean()
print("Probability of Profit:", round(prob_profit, 4))
print("Probability of Loss:", round(prob_loss, 4))
print(
    "Probability of Discount > 20%:",
    round(prob_high_discount, 4)
)
```

16. DECISION MAKING UNDER UNCERTAINTY

```
print("\nDECISION MAKING UNDER UNCERTAINTY")
print("="*50)
# Illustrative business decision table
```

```
decision_data = pd.DataFrame({
    "Decision": [
        "Increase Marketing",
        "Maintain Marketing",
        "Reduce Marketing"
    ],
    "High_Demand": [50000, 35000, 15000],
    "Medium_Demand": [25000, 30000, 18000],
    "Low_Demand": [-10000, 10000, 12000]
})
print("Illustrative Payoff Table:")
display(decision_data)
probabilities = {
    "High_Demand": 0.30,
    "Medium_Demand": 0.50,
    "Low_Demand": 0.20
}
decision_data["Expected_Value"] = (
    decision_data["High_Demand"] *
    probabilities["High_Demand"]
    + decision_data["Medium_Demand"] *
    probabilities["Medium_Demand"]
    + decision_data["Low_Demand"] *
    probabilities["Low_Demand"]
)
display(decision_data)
print(
    "Expected values are calculated using assumed business
    probabilities."
)
```

17. SAMPLING

```
print("\nSAMPLING AND SAMPLING DISTRIBUTION")
print("="*50)
np.random.seed(42)
sample_means = []
for i in range(1000):
    sample = df["Sales"].sample(
        n=50,
        replace=True
    )
    sample_means.append(sample.mean())
sample_means = np.array(sample_means)
print("Number of Samples:", len(sample_means))
print("Mean of Sample Means:", sample_means.mean())
print("Standard Deviation:", sample_means.std())
plt.figure(figsize=(8,5))
plt.hist(sample_means, bins=30)
plt.title("Sampling Distribution of Mean Sales")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.show()
```

18. CONFIDENCE INTERVAL

```

print("\nCONFIDENCE INTERVAL ESTIMATION")
print("="*50)
sales = df["Sales"]
mean_sales = sales.mean()
std_sales = sales.std()
n = len(sales)
for confidence in [0.90, 0.95, 0.99]:
    alpha = 1 - confidence
    t_value = stats.t.ppf(
        1 - alpha/2,
        df=n-1
    )
    margin = t_value * std_sales / np.sqrt(n)
    lower = mean_sales - margin
    upper = mean_sales + margin
    print(
        f"{int(confidence*100)}% CI: "
        f"({lower:.2f}, {upper:.2f})"
    )

```

19. BOOTSTRAPPING

```

print("\nBOOTSTRAPPING")
print("="*50)
np.random.seed(42)
bootstrap_means = []
for i in range(1000):
    sample = np.random.choice(
        df["Sales"],
        size=len(df),
        replace=True
    )
    bootstrap_means.append(
        np.mean(sample)
    )
lower_boot = np.percentile(
    bootstrap_means,
    2.5
)
upper_boot = np.percentile(
    bootstrap_means,
    97.5
)
print("Bootstrap Mean:", np.mean(bootstrap_means))
print(
    "95% Bootstrap CI:",
    (lower_boot, upper_boot)
)
plt.figure(figsize=(8,5))
plt.hist(bootstrap_means, bins=30)
plt.title("Bootstrap Distribution of Mean Sales")
plt.xlabel("Mean Sales")
plt.ylabel("Frequency")
plt.show()

```

20. HYPOTHESIS TESTING

```

print("\nHYPOTHESIS TESTING")
print("="*50)
# H0: Mean profit = 0
# H1: Mean profit != 0
t_stat, p_value = stats.ttest_1samp(
    df["Profit"],
    0
)
print("T-statistic:", t_stat)
print("P-value:", p_value)
alpha = 0.05

if p_value < alpha:
    print(
        "Result: Reject H0. Mean profit is significantly different
        from zero."
    )
else:
    print(
        "Result: Fail to reject H0."
    )

```

21. A/B TESTING

```

print("\nA/B TESTING")
print("="*50)
# Simulated example because original dataset
# does not contain randomized A/B groups
group_A_visitors = 5000
group_B_visitors = 5000
group_A_conversions = 350
group_B_conversions = 410
conversion_A = (
    group_A_conversions /
    group_A_visitors
)
conversion_B = (
    group_B_conversions /
    group_B_visitors
)
print("Group A Conversion Rate:", conversion_A)
print("Group B Conversion Rate:", conversion_B)
successes = np.array([
    group_A_conversions,
    group_B_conversions
])
observations = np.array([
    group_A_visitors,
    group_B_visitors
])
z_stat, p_value = proportions_ztest(
    successes,
    observations
)
print("Z-statistic:", z_stat)

```

```

print("P-value:", p_value)
if p_value < 0.05:
    print(
        "The difference between the two conversion rates is
statistically significant."
    )
else:
    print(
        "The difference is not statistically significant."
    )

```

22. TIME SERIES ANALYSIS

```

print("\nTIME SERIES ANALYSIS")
print("="*50)
monthly_sales_df = (
    df.groupby("Order Month")["Sales"]
    .sum()
    .reset_index()
)
monthly_sales_df["Order Month"] = pd.to_datetime(
    monthly_sales_df["Order Month"]
)
monthly_sales_df = monthly_sales_df.set_index(
    "Order Month"
)
monthly_sales_series = monthly_sales_df["Sales"]
plt.figure(figsize=(12,5))
plt.plot(
    monthly_sales_series
)
plt.title("Monthly Sales Time Series")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.show()

```

23. EXPONENTIAL SMOOTHING FORECAST

```

print("\nEXPONENTIAL SMOOTHING FORECAST")
print("="*50)
model_es = ExponentialSmoothing(
    monthly_sales_series,
    trend="add",
    seasonal=None
).fit()
forecast_es = model_es.forecast(6)
print("Next 6 Month Forecast:")
display(forecast_es)
plt.figure(figsize=(12,5))
plt.plot(
    monthly_sales_series,
    label="Actual"
)
plt.plot(
    forecast_es,
    label="Forecast"

```

```

)
plt.title("Exponential Smoothing Forecast")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.legend()
plt.show()

```

24. ARIMA FORECAST

```

print("\nARIMA FORECAST")
print("="*50)
arima_model = ARIMA(
    monthly_sales_series,
    order=(1,1,1)
).fit()
arima_forecast = arima_model.forecast(
    steps=6
)
print("ARIMA Forecast:")
display(arima_forecast)
plt.figure(figsize=(12,5))
plt.plot(
    monthly_sales_series,
    label="Actual"
)
plt.plot(
    arima_forecast,
    label="ARIMA Forecast"
)
plt.title("ARIMA Sales Forecast")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.legend()
plt.show()

```

25. FORECAST ACCURACY

```

print("\nFORECAST ACCURACY")
print("="*50)

# Hold out last 6 months
train = monthly_sales_series.iloc[:-6]
test = monthly_sales_series.iloc[-6:]
accuracy_model = ExponentialSmoothing(
    train,
    trend="add",
    seasonal=None
).fit()
prediction = accuracy_model.forecast(6)
mae = np.mean(
    np.abs(test.values - prediction.values)
)
mse = np.mean(
    (test.values - prediction.values) ** 2
)
rmse = np.sqrt(mse)

```

```

print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)

# 26. OPTIMIZATION MODELING
print("\nOPTIMIZATION MODELING")
print("="*50)

# Illustrative product allocation problem
# Maximize profit from three product groups
profit_per_unit = np.array([
    40, # Furniture
    30, # Office Supplies
    50  # Technology
])
# Resource constraints
A = np.array([
    [2, 1, 3], # Labour
    [3, 2, 4]  # Storage
])

b = np.array([
    100,
    150
])

# linprog minimizes, therefore use negative profits
result = linprog(
    -profit_per_unit,
    A_ub=A,
    b_ub=b,
    bounds=[(0, None)] * 3,
    method="highs"
)
print("Optimization Successful:", result.success)
if result.success:
    print(
        "Optimal Units:"
    )
    print(
        "Furniture:",
        result.x[0]
    )
    print(
        "Office Supplies:",
        result.x[1]
    )
    print(
        "Technology:",
        result.x[2]
    )
    print(
        "Maximum Estimated Profit:",
        -result.fun

```

```

)

# 27. SIMULATION MODELING
print("\nSIMULATION MODELING")
print("="*50)
np.random.seed(42)
days = 30
daily_demand = np.random.poisson(
    lam=20,
    size=days
)
starting_inventory = 100
inventory = starting_inventory
stockout_days = 0
ending_inventory = []
for demand in daily_demand:
    inventory -= demand
    if inventory < 0:
        stockout_days += 1
        inventory = 0
    ending_inventory.append(
        inventory
    )
print("Total Demand:", daily_demand.sum())
print("Stockout Days:", stockout_days)
print("Ending Inventory:", inventory)
plt.figure(figsize=(10,5))
plt.plot(
    ending_inventory,
    marker="o"
)
plt.title("Inventory Simulation")
plt.xlabel("Day")
plt.ylabel("Inventory")
plt.show()

# 28. ANOVA
print("\nANOVA")
print("="*50)
anova_model = ols(
    "Profit ~ C(Category)",
    data=df
).fit()
anova_table = sm.stats.anova_lm(
    anova_model,
    typ=2
)
display(anova_table)
p_anova = anova_table["PR(>F)"][0]
print("ANOVA P-value:", p_anova)
if p_anova < 0.05:
    print(
        "There is a statistically significant difference in profit
        among categories."

```

```
)
else:
    print(
        "No statistically significant difference was detected
among categories."
    )
```

29. EXPERIMENTAL DESIGN

```
print("\nEXPERIMENTAL DESIGN")
print("="*50)
print("""
Experimental Design:
Factor:
Promotion Type
Levels:
1. Email Marketing
2. Social Media Marketing
Response Variable:
Sales
Experimental Units:
Customers
Procedure:
1. Randomly divide customers into two groups.
2. Apply Email Marketing to Group A.
3. Apply Social Media Marketing to Group B.
4. Record sales.
5. Compare average sales.
6. Perform statistical testing.
Purpose:
To identify whether promotion type affects sales. """)
```

30. FACTORIAL EXPERIMENT

```
print("\nFACTORIAL EXPERIMENT")
print("="*50)
np.random.seed(42)
factorial_data = pd.DataFrame({
    "Promotion": np.repeat(
        ["Email", "Social"],
        40
    ),
    "Discount_Level": np.tile(
        np.repeat(["Low", "High"], 20),
        2
    ),
    "Sales": np.random.normal(
        500,
        100,
        80
    )
})
print("2 × 2 Factorial Experiment:")
display(factorial_data.head())
factorial_model = ols(
    "Sales ~ C(Promotion) * C(Discount_Level)",
```

```
data=factorial_data
).fit()
factorial_anova = sm.stats.anova_lm(
    factorial_model,
    typ=2
)
display(factorial_anova)
```

31. ADDITIONAL BUSINESS ANALYSIS

```
print("\nADDITIONAL BUSINESS ANALYSIS")
print("="*50)
# Average Order Value
average_order_value = (
    total_sales / total_orders
)
print(
    "Average Order Value:",
    round(average_order_value, 2)
)

# Overall Profit Margin
overall_profit_margin = (
    total_profit / total_sales * 100
)
print(
    "Overall Profit Margin:",
    round(overall_profit_margin, 2),
    "%"
)

# Category Profit
category_profit = (
    df.groupby("Category")["Profit"]
    .sum()
    .sort_values(ascending=False)
)
print("\nCategory Profit:")
display(category_profit)

# Segment Sales
segment_sales = (
    df.groupby("Segment")["Sales"]
    .sum()
    .sort_values(ascending=False)
)
print("\nSales by Segment:")
display(segment_sales)
# Sub-category Sales
subcategory_sales = (
    df.groupby("Sub-Category")["Sales"]
    .sum()
    .sort_values(ascending=False)
)
print("\nSales by Sub-Category:")
```

```
display(subcategory_sales.head(10))
```

32. DISCOUNT ANALYSIS

```
print("\nDISCOUNT ANALYSIS")
print("="*50)
df["Discount Band"] = pd.cut(
    df["Discount"],
    bins=[
        -0.01,
        0.10,
        0.20,
        0.30,
        1.00
    ],
    labels=[
        "Low",
        "Medium",
        "High",
        "Very High"
    ]
)
discount_analysis = (
    df.groupby("Discount Band", observed=False)
    .agg(
        Sales=("Sales", "sum"),
        Profit=("Profit", "sum"),
        Average_Profit=("Profit", "mean")
    )
)
display(discount_analysis)
plt.figure(figsize=(8,5))
discount_analysis["Profit"].plot(
    kind="bar"
)
plt.title("Profit by Discount Band")
plt.xlabel("Discount Band")
plt.ylabel("Profit")
plt.xticks(rotation=0)
plt.show()
```

33. BUSINESS INSIGHTS

```
print("\nBUSINESS INSIGHTS AND FINDINGS")
print("="*70)
best_category = category_sales.idxmax()
best_region = region_profit.idxmax()
best_segment = segment_sales.idxmax()
best_subcategory = subcategory_sales.idxmax()
print(
    "1. Highest sales category:",
    best_category
)
print(
    "2. Highest profit region:",
    best_region
```

```

)
print(
    "3. Highest sales customer segment:",
    best_segment
)
print(
    "4. Highest sales sub-category:",
    best_subcategory
)
print(
    "5. Total sales:",
    round(total_sales, 2)
)
print(
    "6. Total profit:",
    round(total_profit, 2)
)
print(
    "7. Total orders:",
    total_orders
)
print(
    "8. Total customers:",
    total_customers
)
print(
    "9. Overall profit margin:",
    round(overall_profit_margin, 2),
    "%"
)
)
```

34. BUSINESS RECOMMENDATIONS

```
print("\nBUSINESS RECOMMENDATIONS")
print("="*70)
print("""
1. Focus on high-performing product categories.
2. Monitor regional performance and improve weaker regions.
3. Track discounts carefully because excessive discounts can
reduce profitability.
4. Use monthly sales trends for inventory planning.
5. Use forecasting models to support future sales planning.
6. Use customer segmentation to create targeted marketing.
7. Use A/B testing before implementing large marketing
campaigns.
8. Use optimization models for better allocation of limited
resources.
9. Use simulation for inventory and demand planning.
10. Regularly monitor KPIs using a BI dashboard. """)
```

35. FINAL PROJECT SUMMARY

```
print("\n")
print("="*70)
print("FINAL BUSINESS ANALYTICS PROJECT
SUMMARY")
print("="*70)
```

```
summary = pd.DataFrame({
    "Metric": [
        "Total Records",
        "Total Sales",
        "Total Profit",
        "Total Quantity",
        "Total Orders",
        "Total Customers",
        "Average Order Value",
        "Overall Profit Margin",
        "Best Sales Category",
        "Best Profit Region",
        "Best Sales Segment"
    ],
    "Value": [
        len(df),
        round(total_sales, 2),
        round(total_profit, 2),
        total_quantity,
        total_orders,
        total_customers,
        round(average_order_value, 2),
        round(overall_profit_margin, 2),
        best_category,
        best_region,
        best_segment
    ]
})
display(summary)
```

- Optimization
- Simulation
- ANOVA
- Experimental Design
- Factorial Experiment

The analysis provides useful business information about sales, profit, customers, products, regions, discounts and future trends.

The results can support better decision-making, marketing, inventory planning and resource allocation. """

```
print("="*70)
print("PROJECT COMPLETED SUCCESSFULLY")
print("="*70)
print("KHUSHI PATEL")
```

36. CONCLUSION

```
print("\nCONCLUSION")
print("="*70)
print("""This Business Analytics project analyzed Superstore
sales data using Python and statistical techniques.
```

The project covered:

- Data Collection
- Data Preprocessing
- Exploratory Data Analysis
- Descriptive Statistics
- Distribution Analysis
- Correlation Analysis
- Regression Analysis
- Business Intelligence
- Probability
- Decision Making Under Uncertainty
- Sampling
- Confidence Intervals
- Bootstrapping
- Hypothesis Testing
- A/B Testing
- Time Series Analysis
- Forecasting
- Forecast Accuracy

2] Data Set Sample :

DATASET LOADED SUCCESSFULLY
Rows: 10000
Columns: 24

First 5 Records:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Category	Sub-Category	Product Name
0	1	US-2017-000001	01/31/2017	02/03/2017	Same Day	CU-01324	Customer 355	Corporate	United States	Atlanta	...	Technology	Phones	Phones Product 1
1	2	US-2017-000002	12/30/2017	01/02/2018	Standard Class	CU-02530	Customer 2464	Corporate	United States	New York City	...	Office Supplies	Art	Art Product 2
2	3	US-2016-000003	05/10/2016	05/11/2016	Second Class	CU-00137	Customer 708	Consumer	United States	Phoenix	...	Office Supplies	Labels	Labels Product 3
3	4	US-2017-000004	07/18/2017	07/19/2017	Same Day	CU-02793	Customer 2127	Corporate	United States	Dallas	...	Office Supplies	Labels	Labels Product 4
4	5	US-2017-000005	02/04/2017	02/09/2017	Standard Class	CU-00933	Customer 163	Home Office	United States	San Diego	...	Office Supplies	Art	Art Product 5

5 rows x 24 columns

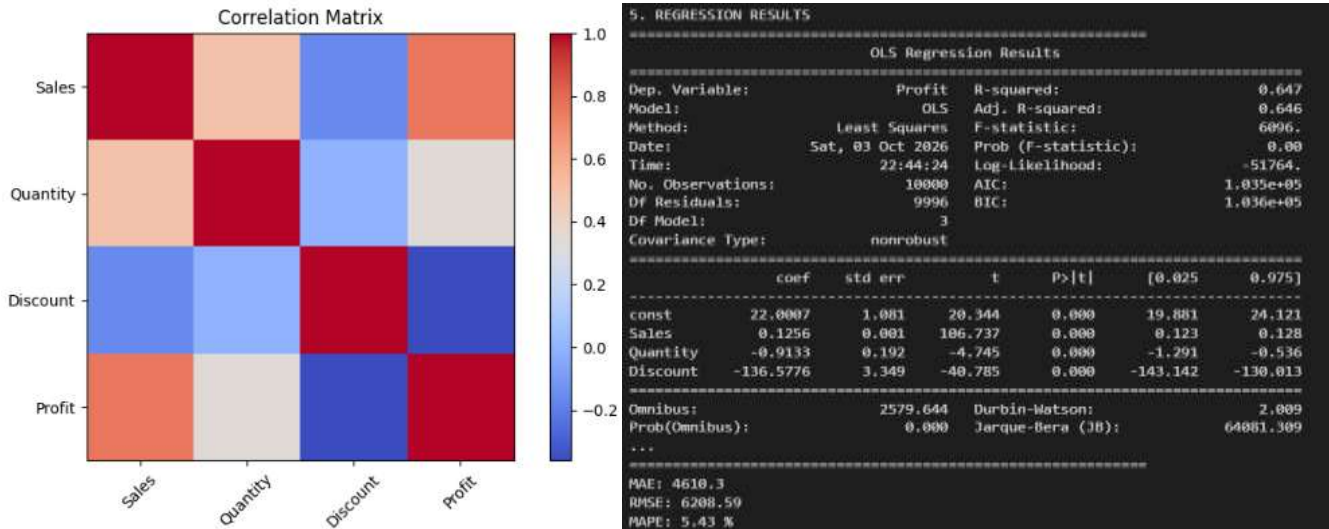
3] Important Outputs :

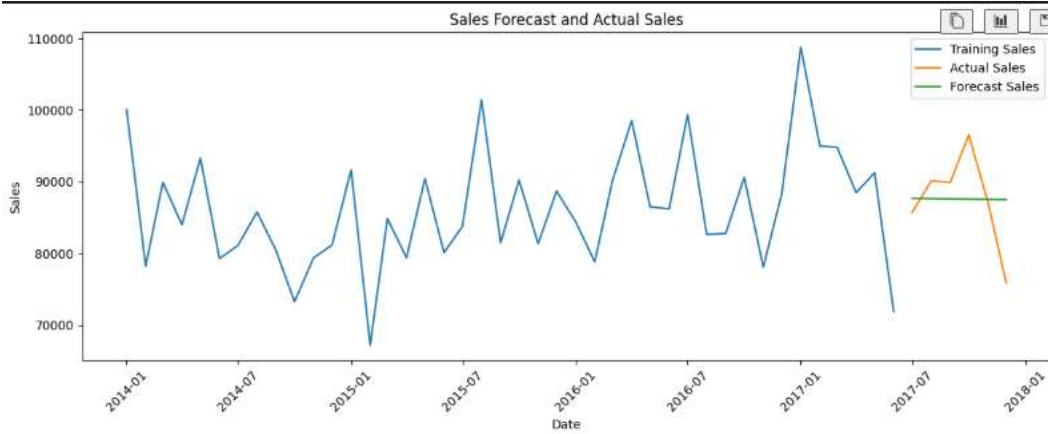
DATASET INFORMATION

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 24 columns):
#   Column              Non-Null Count  Dtype
---  ---
0    Row ID              10000 non-null  int64
1    Order ID            10000 non-null  object
2    Order Date          10000 non-null  object
3    Ship Date           10000 non-null  object
4    Ship Mode           10000 non-null  object
5    Customer ID         10000 non-null  object
6    Customer Name       10000 non-null  object
7    Segment             10000 non-null  object
8    Country             10000 non-null  object
9    City               10000 non-null  object
10   State              10000 non-null  object
11   Postal Code        10000 non-null  int64
12   Region             10000 non-null  object
13   Product ID         10000 non-null  object
14   Category           10000 non-null  object
15   Sub-Category       10000 non-null  object
16   Product Name       10000 non-null  object
```

1. MISSING-VALUE CHECK

```
Row ID      0
Order ID    0
Order Date  0
Ship Date   0
Ship Mode   0
Customer ID  0
Customer Name 0
Segment     0
Country     0
City        0
State       0
Postal Code  0
Region      0
Product ID  0
Category    0
Sub-Category 0
Product Name 0
Sales       0
Quantity    0
Discount    0
Profit      0
Order Month 0
...
Sales      1.000000  0.497440 -0.156189  0.765198
Quantity   0.497440  1.000000 -0.010157  0.339456
Discount   -0.156189 -0.010157  1.000000 -0.361995
Profit      0.765198  0.339456 -0.361995  1.000000
```





- Dataset information
- Missing-value check
- Duplicate check
- Descriptive statistics
- Correlation matrix
- Regression results
- Confidence interval
- Hypothesis testing
- Forecast accuracy
- ANOVA results

9. ANOVA RESULTS

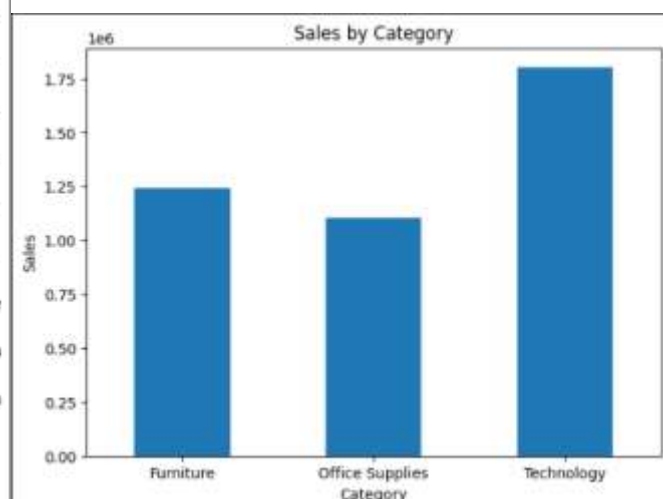
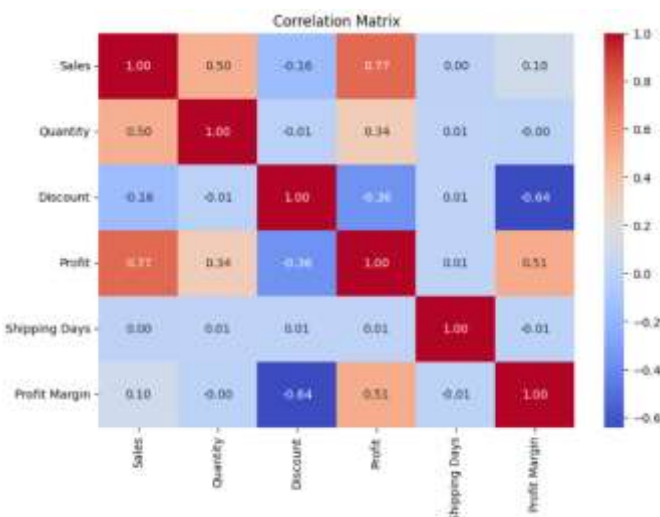
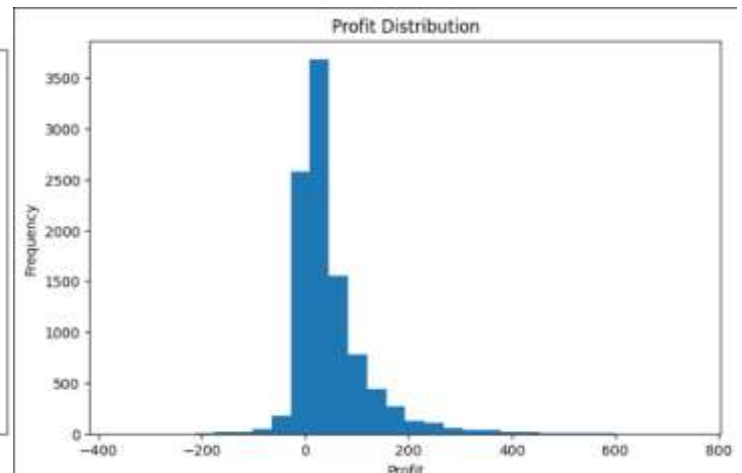
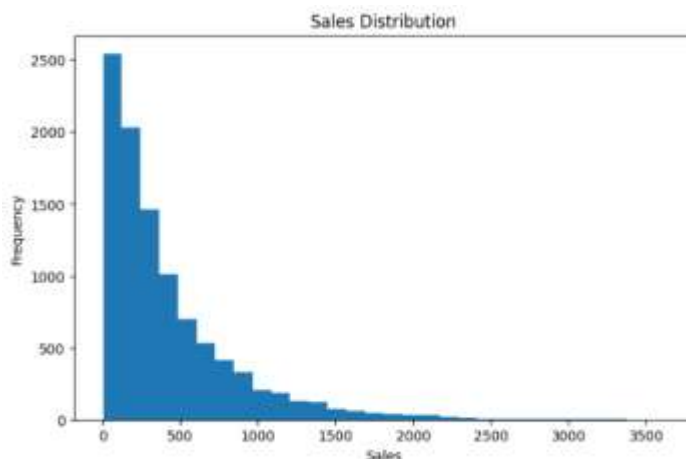
	sum_sq	df	F	PR(>F)
C(Category)	4.439043e+06	2.0	467.212929	1.053425e-194
Residual	4.749131e+07	9997.0	NaN	NaN

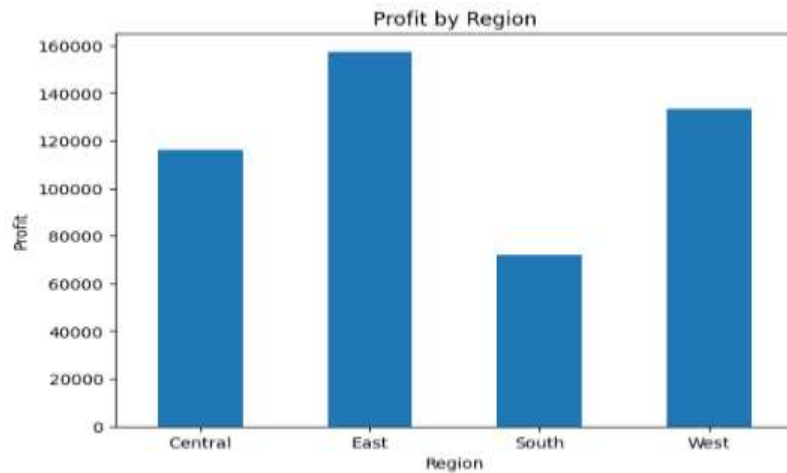
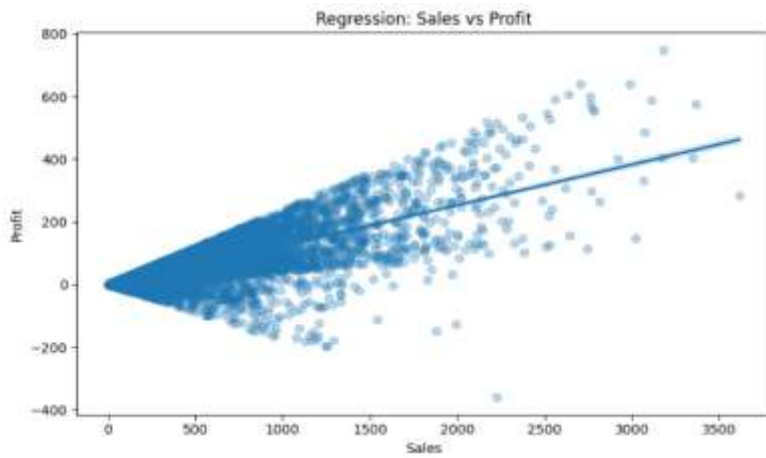
ANOVA P-Value: 0.0

Decision: Reject the Null Hypothesis.

There is a statistically significant difference in average Profit among categories.

4] Additional Graphs :

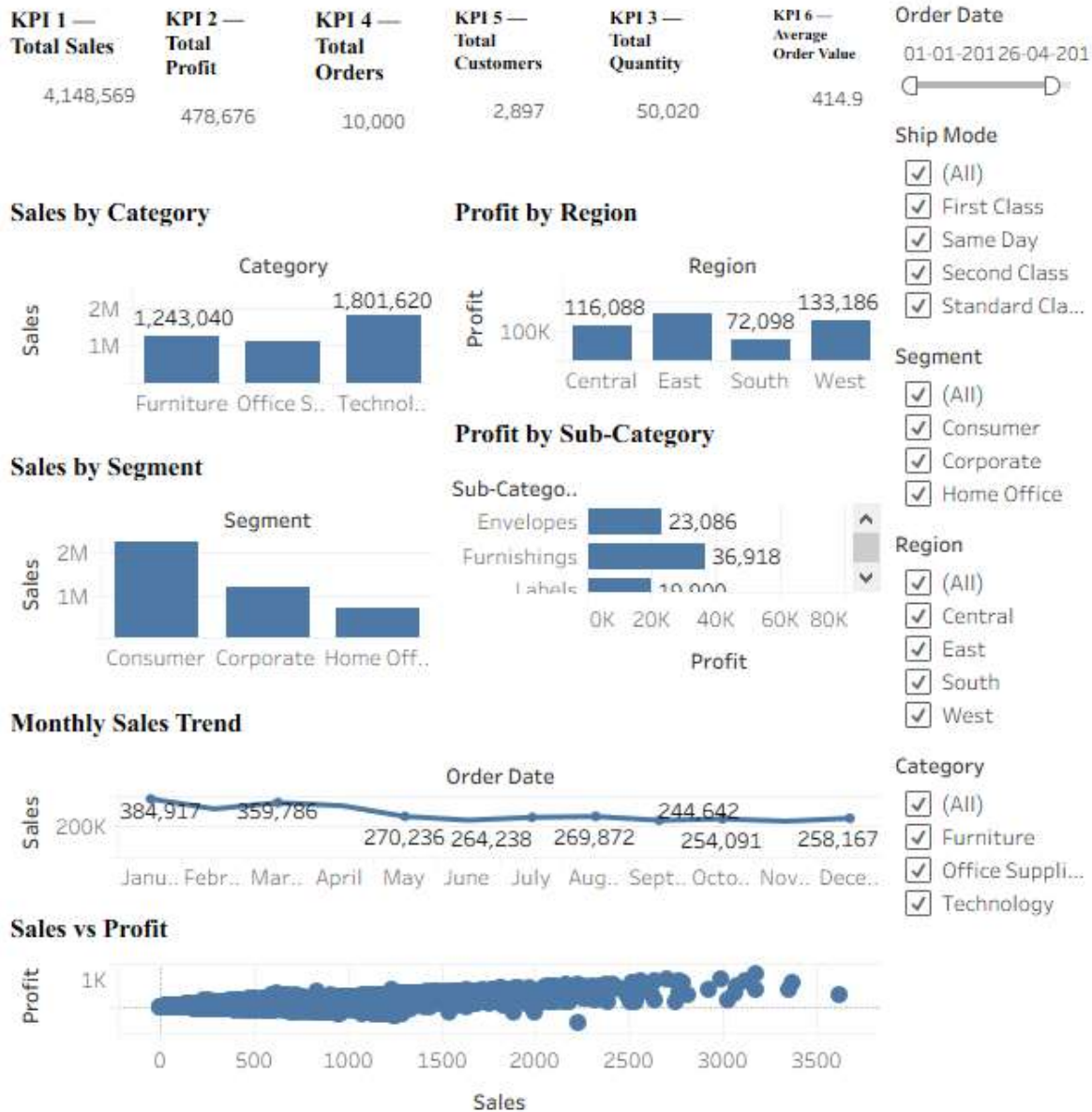




Sales distribution

1. Profit distribution
2. Correlation heatmap
3. Sales by category
4. Profit by region
5. Monthly sales trend
6. Regression/scatter plot
7. Forecast graph

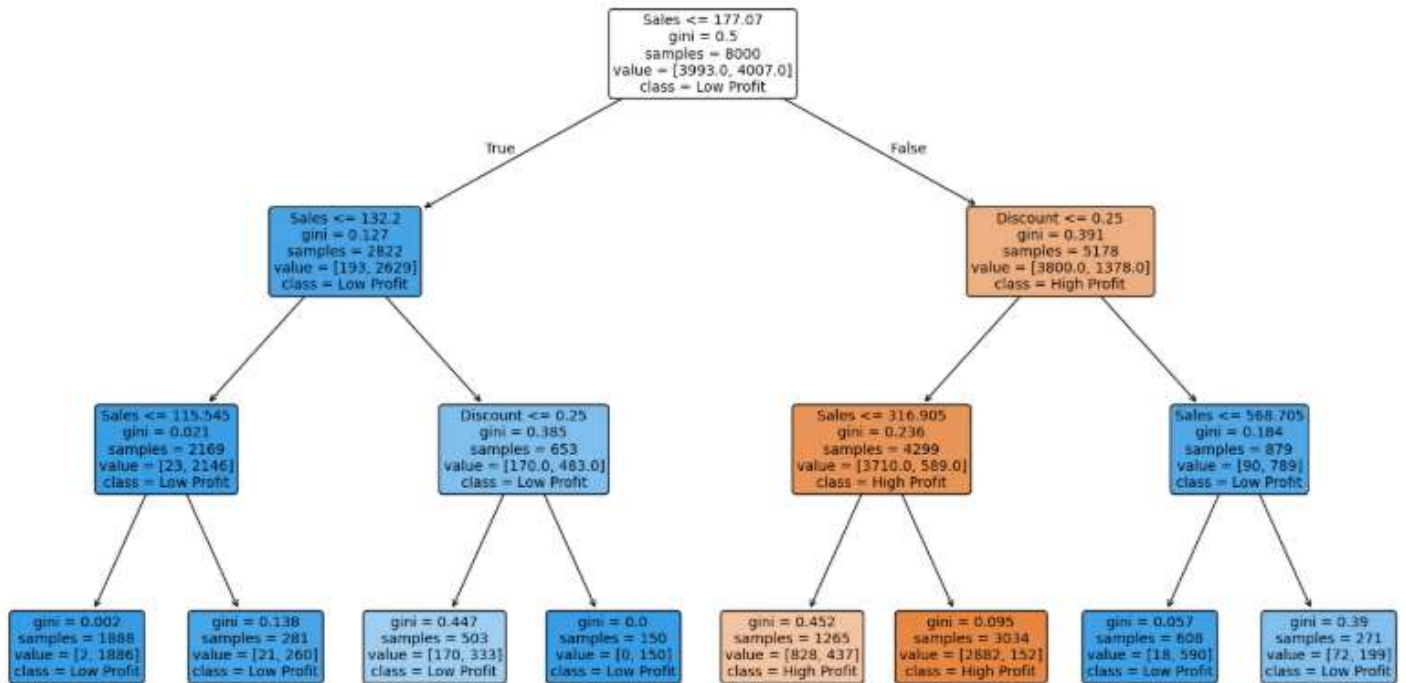
SUPERSTORE BUSINESS ANALYTICS DASHBOARD



Sheet 7 Sheet 8 Sheet 9 Sheet 10 Sheet 11 Sheet 12 Superstore Business Analyt...

6 | Decision Tree :

Decision Tree for Profit-Level Decision Making



7] Optimization Output :

OPTIMIZATION MODELING

Optimization Successful: True

Optimal Units:

Furniture: 0.0

Office Supplies: 75.0

Technology: 0.0

Maximum Estimated Profit: 2250.0

8] Simulation Output :

SIMULATION MODELING

Total Demand: 585

Stockout Days: 25

Ending Inventory: 0

Inventory Simulation

